

Cardiovascular Risk Prediction Among South Asian Populations

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Type of Project:

Machine Learning / Predictive Analytics

Goals Achieved:

- Predict Cardiovascular disease using machine learning
- Determine which algorithms are the most accurate for prediction

Technologies used:

- Python
- Google Colab
- Scikit-learn
- Pandas
- Matplotlib
- Seaborn

Dataset Used:

“Cardiovascular Disease Risk Assessment Dataset” by Ahmed Mohamed Zaki

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Abstract

Cardiovascular disease (CVD) is a major health concern among all populations. Interestingly though, South Asians have a disproportionately higher risk of CVD. This project investigates the causes of CVD and uses machine learning techniques to predict the chances of developing CVD. Different factors such as demographics, clinical, and lifestyle factors are just some of the key causes of cardiovascular disease. The dataset was cleaned and preprocessed to address missing values, inconsistent data, outliers, and categorical variables. Using exploratory analysis and statistical methods such as the Kruskal-Wallis test, relationships between predictors and CVD risk levels were studied to determine any statistical significance and how they might impact patients. Multiple classification models such as Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, and XGBoost were trained and evaluated using appropriate statistical and classification metrics. The results were compared to determine which model would provide the strongest predictive performance. The findings demonstrated that the Gradient Boosting model had the highest predictive accuracy and was the most precise in terms of predicting the risks for CVD. Despite this however, there are plenty of limitations when it comes to dataset composition, data accuracy, bias, generalizability, and the use of predictive models in healthcare.

Introduction

Background

Cardiovascular diseases (CVDs) are a “group of disorders of the heart and blood vessels” [11]. Some examples of CVDs include coronary heart disease (disease of the blood vessels supplying the heart muscle), cerebrovascular heart disease (disease of the blood vessels supplying the brain), peripheral arterial disease (disease of the blood vessels supplying the arms and legs), and many more [9, 11]. These can lead to serious complications which include, but aren’t limited to; numbness, confusion, strokes, heart failure, heart attacks, and many more [. Knowing about and understanding CVDs are very important because they are the leading cause of death globally, they impact people of all ages, and can cause life-threatening issues if not treated early on. It is extremely important to identify risks early on to ensure proper care is given and that the disease doesn’t get worse with age and lifestyle changes.

South Asian Cardiovascular Risk

South Asian adults start developing the risk for CVD earlier compared to other populations. In a lot of cases, it can occur during their thirties and forties. This is very interesting because South Asians also have overall healthier lifestyle choices ranging from lower alcohol use to a higher diet quality [1]. At the age of 45, South Asian men had an eight-times higher probability of

having prediabetes with women having a three-times higher probability compared to their peers respectively [1]. To further put this into how early South Asians can contract these diseases, South Asian men had a much higher chance of having blood pressure, about 25.5% compared to other population which had below 20%, and 78.2% of South Asian men had a higher risk of having high cholesterol and/or triglycerides [1]. Another issue is that a lot of healthcare professionals tend to undermine how much of these symptoms should be taken seriously in minority groups. Their risk factors include, but are not limited to, higher chances of having prediabetes, hypertriglyceridemia (high levels of triglycerides), increased chances of being overweight and obese, and many more [8]. Since these numbers are disproportionately higher, a population-specific study would be beneficial because it would ensure South Asians are the main focus, it would uncover any unique patterns, and target solutions specifically for that population that is being studied.

Problem Statement

Cardiovascular disease is a major health burden, and South Asian populations experience disproportionately higher cardiovascular risks compared to other populations. This project will investigate whether machine learning can help identify patterns within different patterns such as demographic, clinical, and lifestyle variables in order to predict CVD risk and come up with solutions early on to prevent further damage.

Questions Asked

- Can machine-learning classification models accurately predict cardiovascular disease risk levels among individuals in a South Asian population using demographic, clinical, and lifestyle characteristics?
- Alongside this, which variables are most strongly associated with CVD risk? Which machine learning model provides the best predictive performance?
- Which patient characteristics contribute most to the model predictions?
- Finally, what limitations affect the reliability and generalizability of the predictions?

Project Objectives

Some objectives that have been tackled here are cleaning and preprocessing the CVD dataset and identifying and addressing missing and inconsistent data. An exploratory data analysis must be conducted to understand the variables and the dataset while also studying which predictors would be the most relevant to the models. A statistical analysis is performed for all variables to determine the significance of the features when it comes to predicting the risk of CVD. Then we want to train and compare different classification models. Finally, we want to identify important predictive features and determine the limitations and potential applications of these models.

Dataset

Dataset Source

This dataset was obtained from Kaggle, a public repository of many different datasets. This dataset is titled “Cardiovascular Disease Risk Assessment Dataset”. It is published by Ahmed Mohamed Zaki. It has 22 different lifestyle attributes along with 1,529 complete patient records. This dataset represents people living in Bangladesh which has a very diverse set of people. This dataset, while focusing on people from Bangladesh, can also be used to further support risk assessments in other South Asian countries due to similar conditions. The target variable for this dataset is the CVD Risk Level.

Variable Description

Variable	Type	Description
Age	Numerical	Patient Age
Sex	Categorical	Sex of Patient
BMI	Numerical	Body Mass Index
Smoking Status	Categorical	Does the patient smoke?

Appendix A

Data Preprocessing

Initial Data Inspection

The first things that were inspected in the dataset were the types of data that made up the dataset. The initial dataset included a mix of float64 and object data types for numerical and categorical variables respectively. After that, the amount of data columns, categories, and null values were studied. In the initial dataset, there were 1,529 entries with 22 total data columns. The inspection also revealed a lot of missing (null) values but besides that, every other piece of data was consistent in terms of validity, reliability, and data cleanliness [Appendix B].

Missing Data Analysis

In the dataset, all of the float64 data types had missing values. They ranged from 69 missing values to 82 missing values. The variable that had the most missing values was the “Diastolic BP” variable with 82 missing values. Instead of dropping these values completely, it was best if the data was imputed. This is because if the missing data was dropped, the sample size would’ve been greatly reduced, there would be bias from data loss, and having all of the features in a patient ensures the most accurate models when they get trained. For the numerical variables,

using mean imputation worked the best because these variables have numbers that can be added and average which would be impossible to do in categorical variables since they are text-based. For categorical variables, using mode imputation worked the best because mode is finding the most frequent values which doesn't require numbers to do that.

Handling Inconsistent Data

There were no cases of inconsistent data from the original dataset. Manually studying the dataset and further studying it using code showed that all data was consistent with just the missing values which have already been accounted for previously.

Outlier Detection

Outliers needed to be identified if there were any because they can skew the statistical distributions, distort calculations, cause models to overestimate and overfit, and degrade overall prediction accuracy. In order to identify outliers, this code was run and after some thorough inspection, it was noticed that the Waist-to-Height Ratio Distribution and the CVD Risk Score Distribution had outliers to the right of the boxplots. Since removing values would impact the accuracy of the model and the fact that this model is also based on real patients, it was decided that these outliers wouldn't be removed from the dataset.

Derived Variables

A lot of variables had missing values. These values were filled up through estimations, masks, and statistical imputations. For the Height variables (cm & m) and BMI, masks were used to filter the rows that need to be manipulated. If the height in one row is missing, if there is another row with the converted version of the missing height, then it needs to either multiply or divide by 100 in order to get the missing values. This rule also applies to BMI but instead the missing value is discovered by dividing the Weight (kg) and the square of the Height (m). The rest of the variables were found using the mean imputations since those are all individual variables. For categorical variables, they were filled using the mode which takes the most frequent object in each row and implements them into the missing values. Finally, the target column ("CVD Risk Level") has its missing values handled by simply dropping them since these do not have a large impact on the raw dataset.

Categorical Encoding

In order for the machine learning models to perform calculations, give predictor scores, and avoid data leakage, the categorical variables must be converted into numeric values using a method called one-hot encoding. Using a mix of zeroes and ones, this will allow us to implement these variables in the machine learning models in order to perform the necessary calculations needed. First a label encoding tool was performed into all of our categorical variables which include; Sex, Smoking Status, Diabetes Status, Physical Activity Level, Family History of CVD, Blood Pressure Category, and CVD Risk Level. This will allow the categorical variables to

convert into numerical integers ranging from 0-n1 [4]. First the risk levels were encoded with 2 representing “LOW” risk, 1 representing “INTERMEDIARY” risk, and 0 representing “HIGH” risk. For the Sex variable, 0 represents F (female) and 1 represents M (male). For values that have yes and no values such as “Family History of CVD”, 0 represents NO and 1 represents YES.

Train-Test Split

The dataset was divided into training and testing subsets before the models were trained. The training was used to fit the model whereas the testing set was reserved for evaluating the ability to generalize to previously unseen observations [Appendix C].

Exploratory Data Analysis

Introduction

The dataset has 22 variables. In order to understand the limitations of our dataset and what the variables mean, an EDA (exploratory data analysis) must be performed. Some variables have some redundancies such as the height variables while others are independent variables. Analyzing the dataset also allows for the identification of potential outliers and relationships. Visualizations are made to better understand the relationship between the target variable and the other variables that are being compared against it. This analysis allows for an opportunity to determine if the cleaned dataset is of high quality and if it is solid enough for model training.

Target Distribution

Performing a line of code that shows the model distribution showed some very interesting results. The distribution of targets is highly imbalanced. The graph shows that there is a significantly lower number of patients with a “LOW” CVD Risk Level compared to patients with “INTERMEDIARY” or “HIGH” risk levels. Approximately 220 people (or about 14.4% of the population) have a “LOW” risk level compared to the ~580 and ~730 people (~37.9% and 14.4% respectively) who have “INTERMEDIARY” and “HIGH” risk levels. There are many possible explanations such as a smaller sample size or what is most likely the case is that the underlying population has more people at risk of CVD. This would most likely be the case as further studies have shown that South Asians tend to have the highest risk of getting CVD or other heart related issues due to a variety of factors like genetics and risk for other diseases like prediabetes.

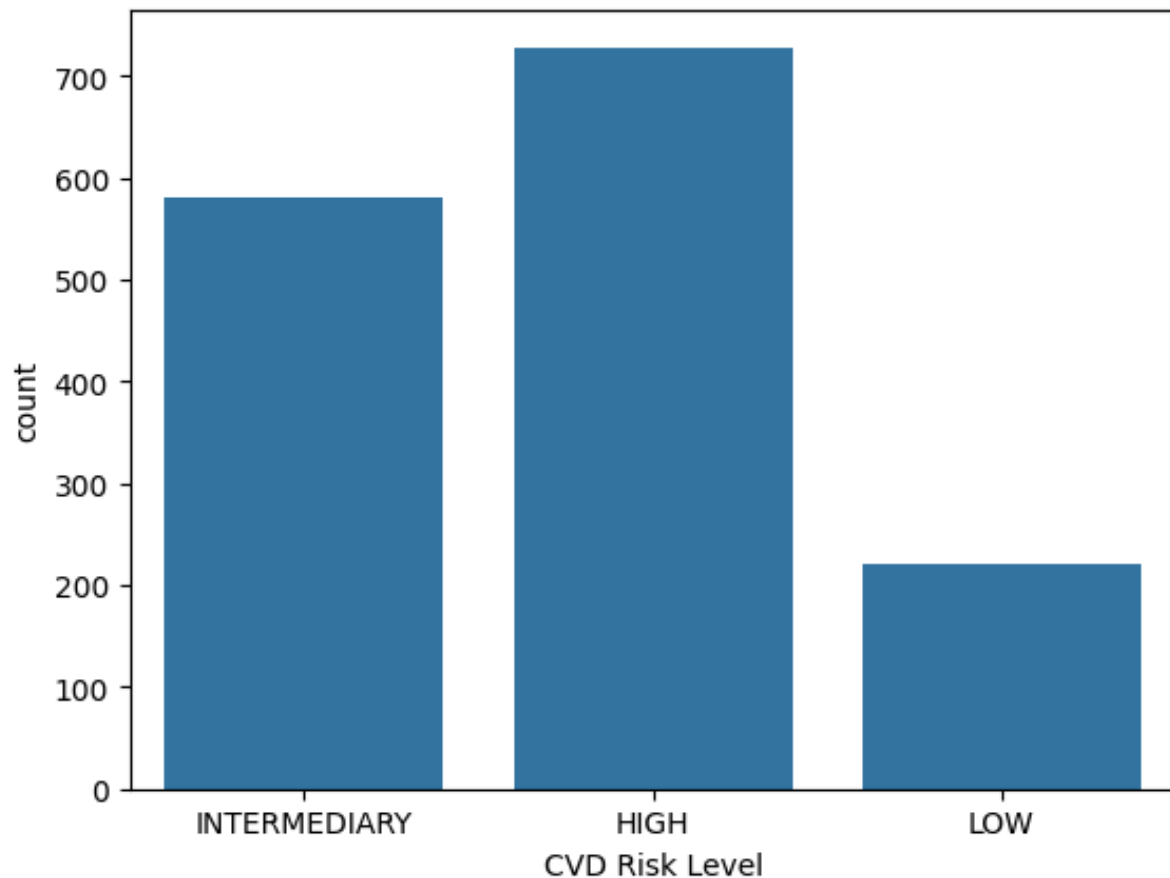


Figure 1: Disease Distribution Countplot by "CVD Risk Level"

Age Distribution

Age is one of the biggest factors when it comes to being at risk for cardiovascular disease. By observing the age histogram and boxplots, it can be determined that the people who get older, their risk of being at risk for CVD or already having CVD generally increases substantially. From the ages 30-60, the "LOW" risk tends to be concentrated at those ages with a few older patients. In the "INTERMEDIARY" risk level, patients ages 30-65 are spread across with a higher concentration on patients in their late 30s/early 40s. Finally the "HIGH" risk levels show patients ranging from ages 45-60. The green curve represents higher risk patients and reaches its peak around age 50.

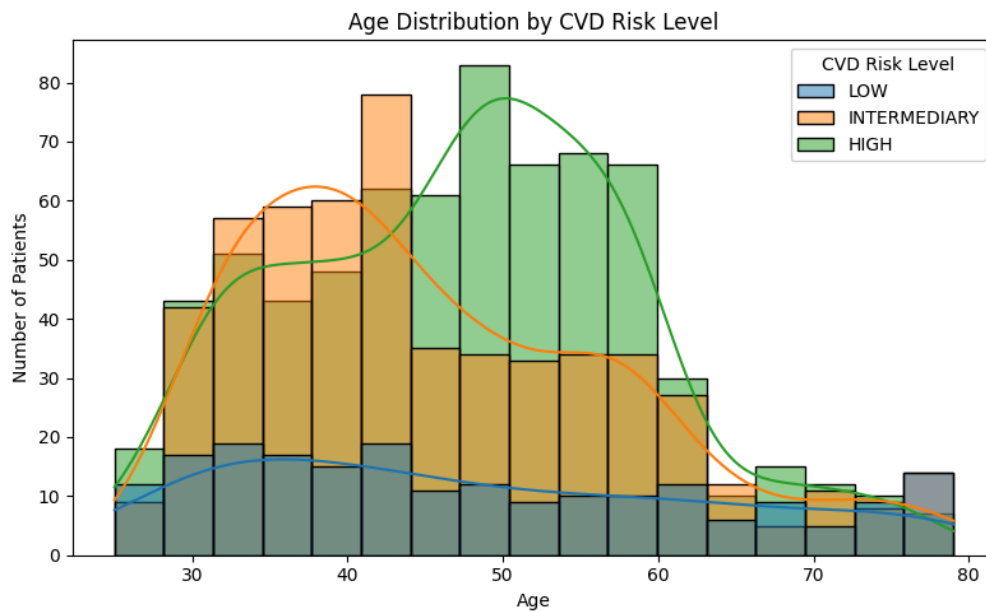


Figure 2: Age Distribution Histogram by CVD Risk Level

BMI Distribution

In the BMI histogram, it is shown that generally, the higher the patients' BMI is, the higher their risk level is. Patients who have a BMI of 15 to ~25 are clustered in the "LOW" and "INTERMEDIARY" risk levels. Patients higher than a BMI of 25 tend to be clustered in the "HIGH" risk levels. There are a significant number of people that have a BMI of 30-40 and those people have the highest risk of getting CVD. Another thing to keep note is that there is a lot of overlap between these results meaning that there are patients that have a higher risk level even though generally, they might be in a lower generalized risk population. Overall though, BMI is not a very strong solo indicator of higher CVD risk as it cannot reliably distinguish between the three CVD risk categories. This can be further shown in the boxplots and it is shown that there are a lot of BMI values that show multiple risk types.

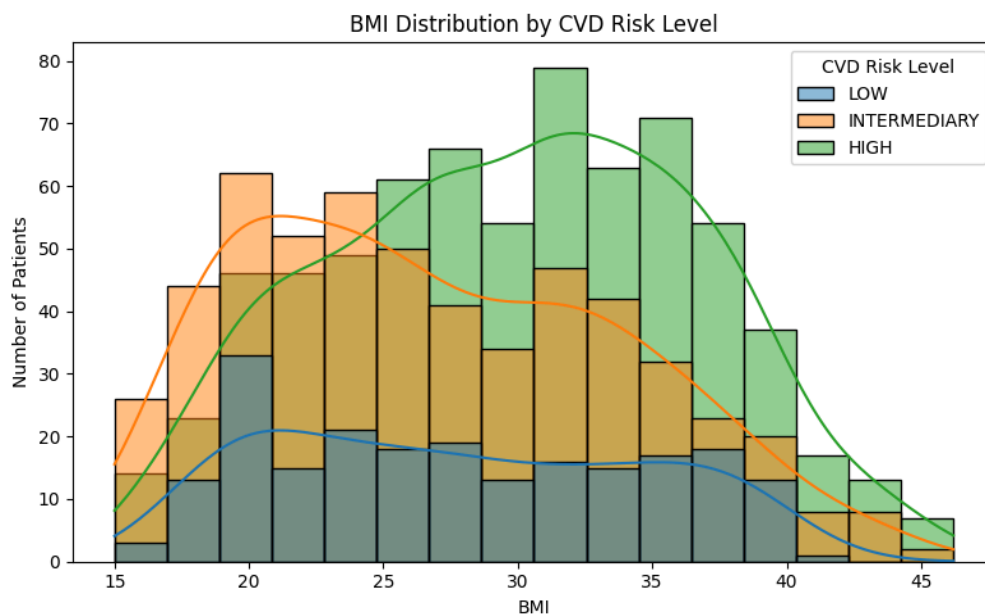


Figure 3: BMI Distribution by CVD Risk Level Histogram

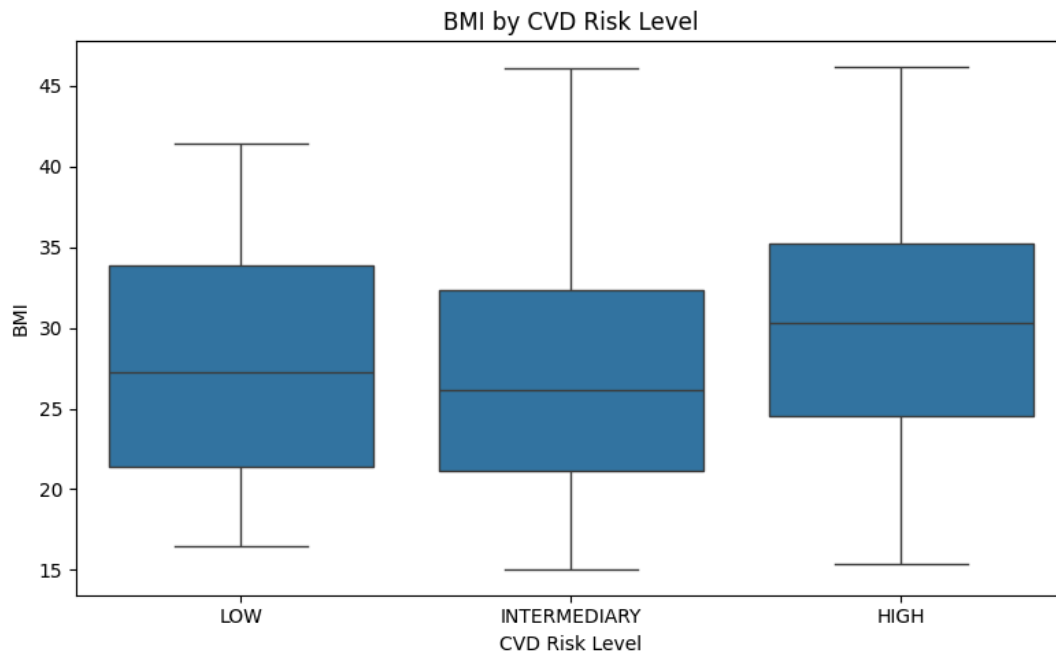


Figure 3a: BMI by CVD Risk Level Boxplot

Fasting Blood Sugar Distribution

In the histogram and boxplots, the biggest takeaway is that generally, patients with higher fasting blood sugar levels tend to have higher risks of CVD. It can be proven by observing the bars which shows that levels ranging from 80-150 mg/dL tend to have “INTERMEDIARY” and “HIGH” risk levels with higher levels being at “HIGH” risk. The green curve peaks at around 95 mg/dL indicating that it is not only blood sugar levels that determine CVD risks. There is also general overlap in both graphs further proving that there are other factors that influence risk levels.

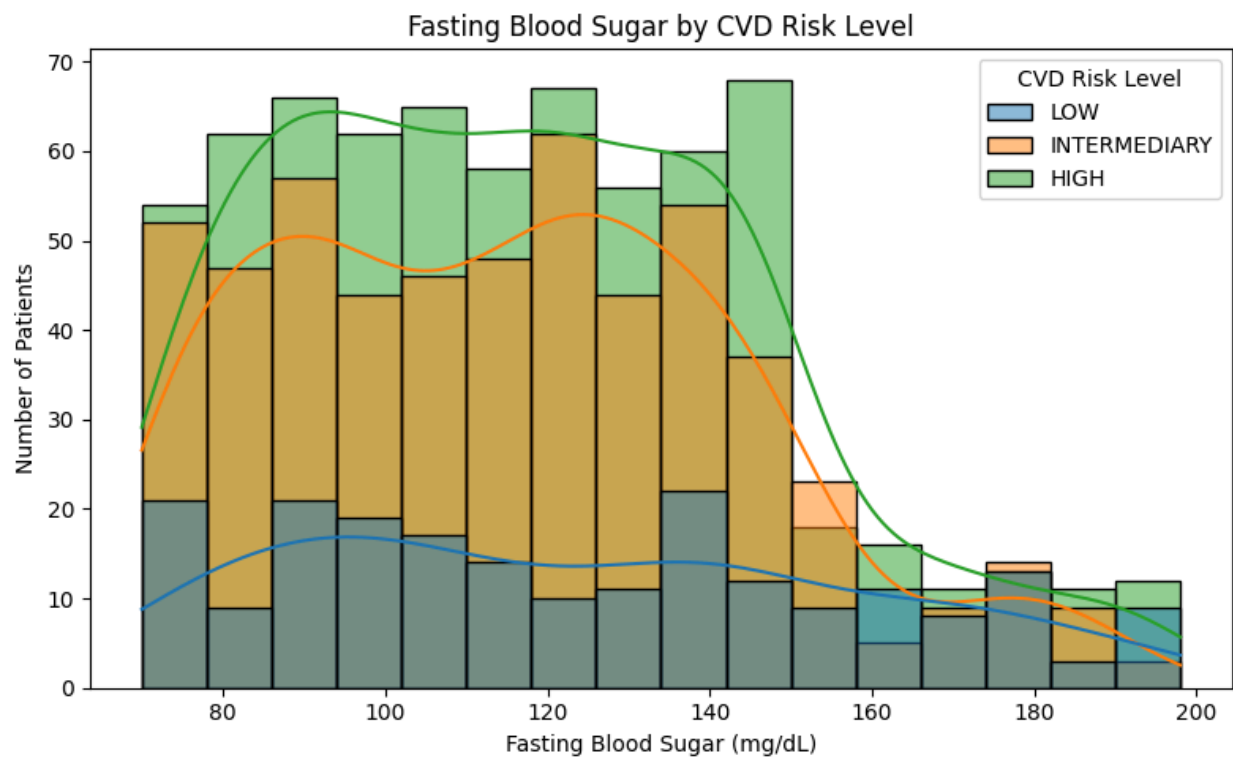


Figure 4: Fasting Blood Sugar by CVD Risk Level Histogram

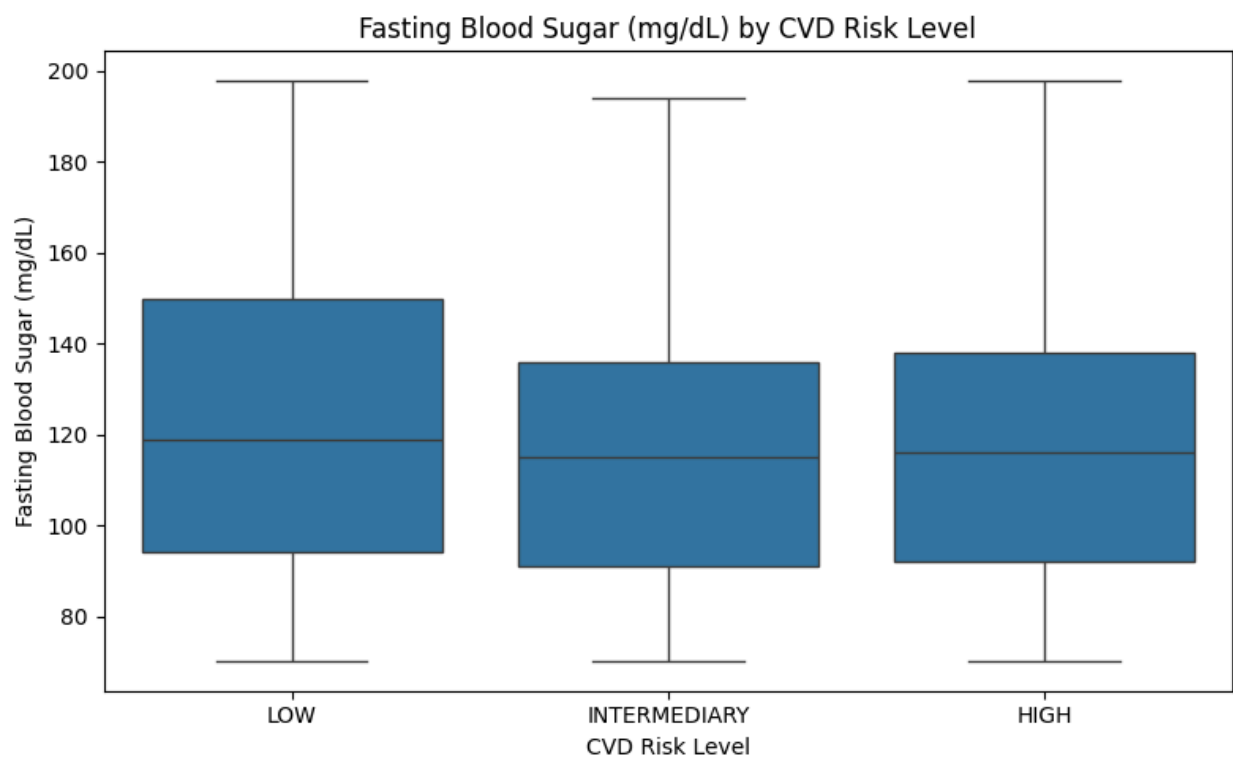


Figure 4a: Fasting Blood Sugar by CVD Risk Level Boxplot

Systolic & Diastolic BP Distribution

The systolic BP boxplot shows that “LOW” risk groups have the highest ranges of Systolic BP ranging from around 115-165 mmHg with a median value of around 130 mmHg which is the opposite of what would be assumed. The patients that have “INTERMEDIARY” and “HIGH” risk levels have slightly lower ranges ranging from around 115-140 mmHg with median values of around 123 mmHg. However, just like the other variables, systolic BP alone won’t accurately predict CVD risk as there is plenty of overlap between the values.

The diastolic BP boxplot shows that the “LOW” risk category actually has the highest median diastolic BP of about 87 mmHg with a range from around 72-98 mmHg. When compared to the “INTERMEDIARY” and “HIGH” risk categories, these ranges are similar but they have slightly lower medians of about 81 and 82 mmHg respectively. This pattern is similar to the systolic distribution but also shows that these two alone are not going to accurately predict CVD risk.

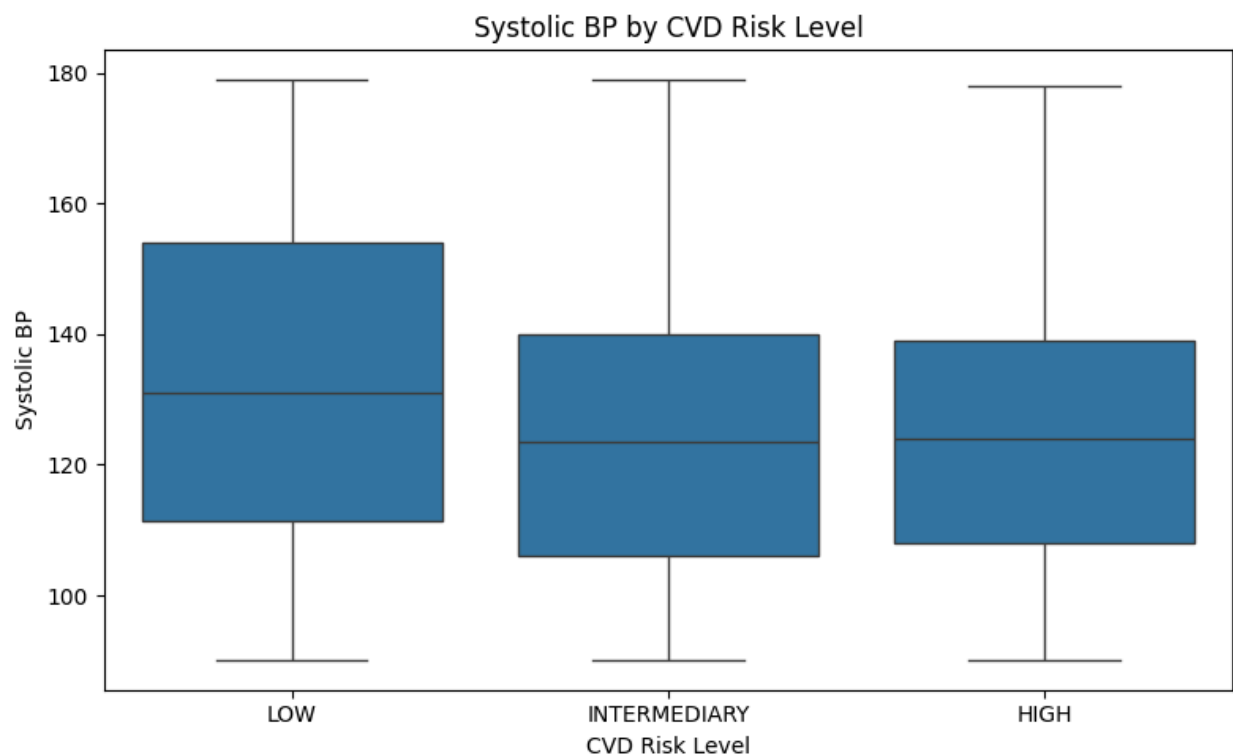


Figure 5: Systolic BP by CVD Risk Level Boxplot

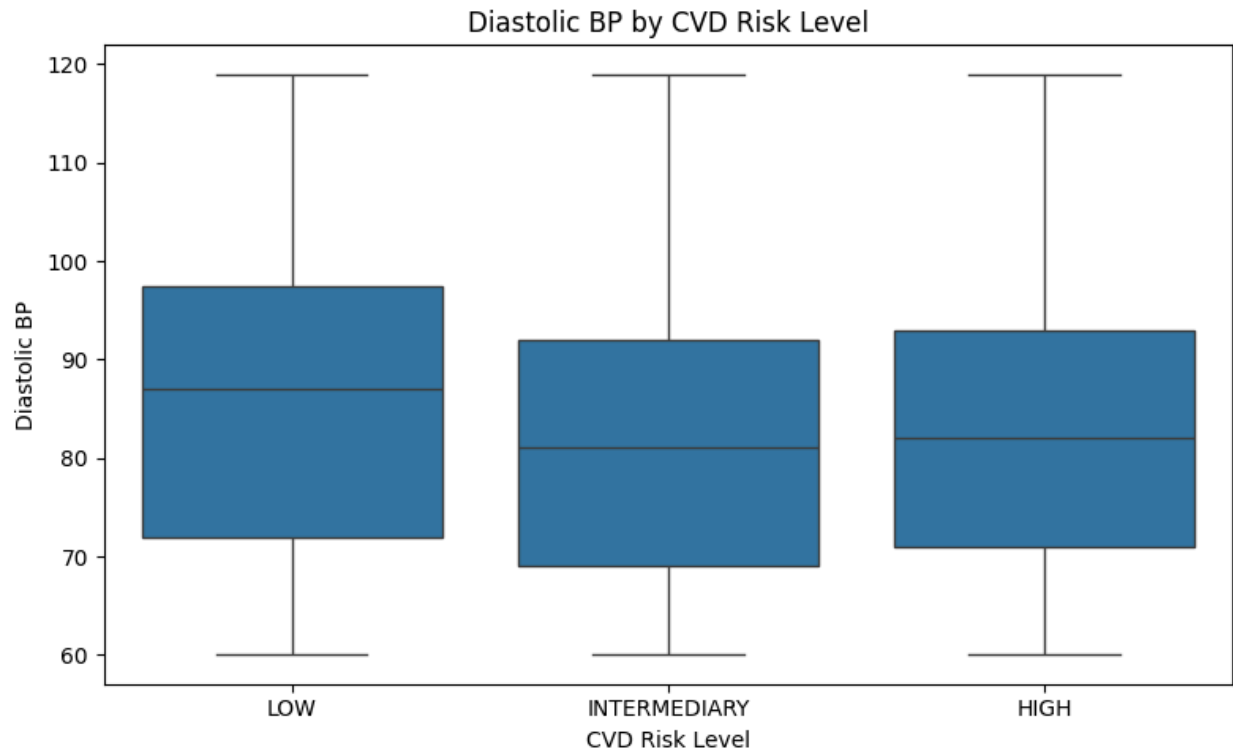


Figure 6: Diastolic BP by CVD Risk Level Boxplot

Sex by CVD Risk Level

When creating and observing the countplot to compare the sex of a person with their risk of getting CVD, it is shown that females tend to have slightly higher amount of patients that have a high chance of getting CVD and this is most likely due to the fact that there are slightly more females than males in the original dataset. However, one thing that is interesting is that both males and females have significantly higher chances of having “HIGH” risk of CVD with 355 females having a high risk and around 350 males having the same risk. This alone wouldn’t be enough to predict CVD outcomes, but it is a very important factor since it can indicate that with other factors, the population is genetically predisposed to cardiovascular risk.

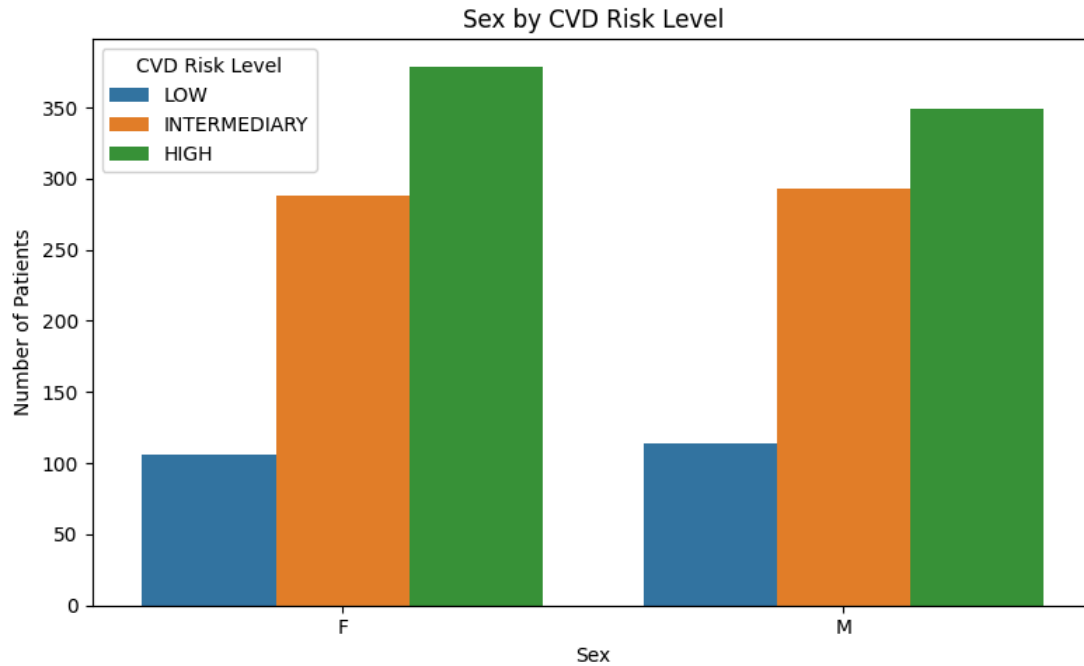


Figure 7: Sex by CVD Risk Levels Countplot

Smoking Status by CVD Risk Level

The countplot to compare smoking status shows some details that were mostly expected. It shows that patients who smoked had a much higher risk of receiving CVD compared to nonsmokers. While this does not mean it is a causation of CVD, it does contribute to a lot of the symptoms that are associated with it. However, nonsmokers still had a majority of patients who were in the “INTERMEDIATE” category for CVD. This indicates that smoking status alone won’t predict CVD outcomes but it is a very strong variable to consider in the model.

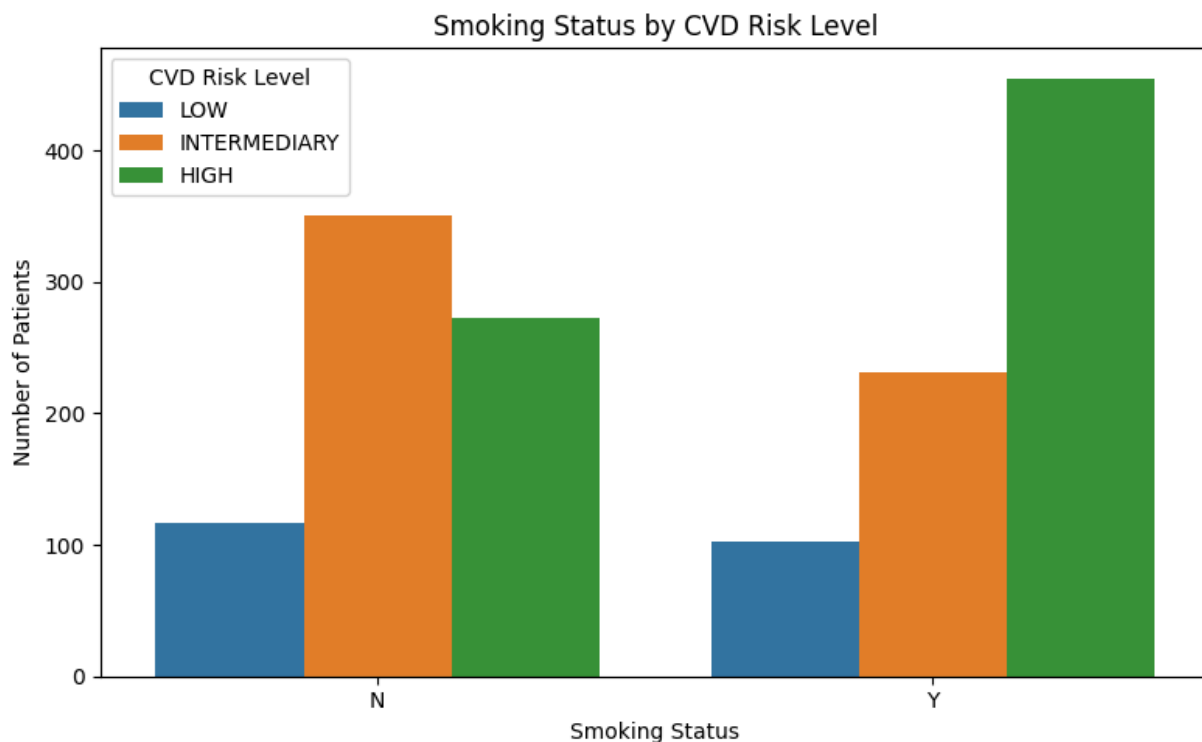


Figure 8: Smoking Status by CVD Risk Level Countplot

Diabetes Status by Risk Level

In this countplot, it is shown that patients that have diabetes generally have the highest risk of also having CVD compared to patients who do not have diabetes. This is expected because one of the main diseases that can also cause CVD is diabetes. Despite that, non-diabetic patients still have a large number of “HIGH” and “INTERMEDIARY” cases which indicates that diabetes alone will not be the sole cause of CVD, but pairing it with other variables, it is a solid variable to consider in the model.

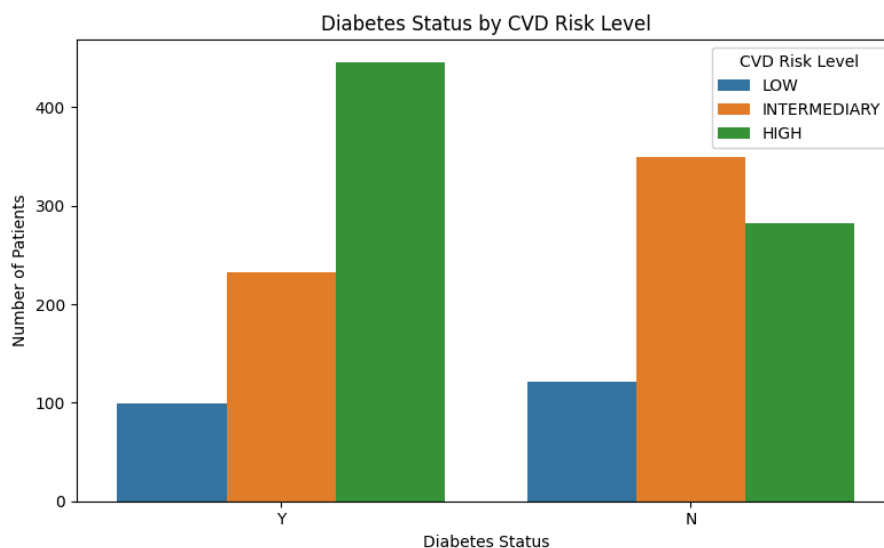


Figure 9: Diabetes Status by CVD Risk Level

Family History by CVD Risk Level

This countplot shows patients who reported having family members with CVD. It is shown in this plot that patients who have/had family members with CVD generally have the highest risk levels of getting CVD compared to those who reported not having family members with CVD. Regardless, just as the other variables before, patients who have reported not having family members with CVD also have the highest “INTERMEDIARY” risk when it comes to getting CVD which indicates that this alone will not predict CVD risk but it is a strong variable to consider.

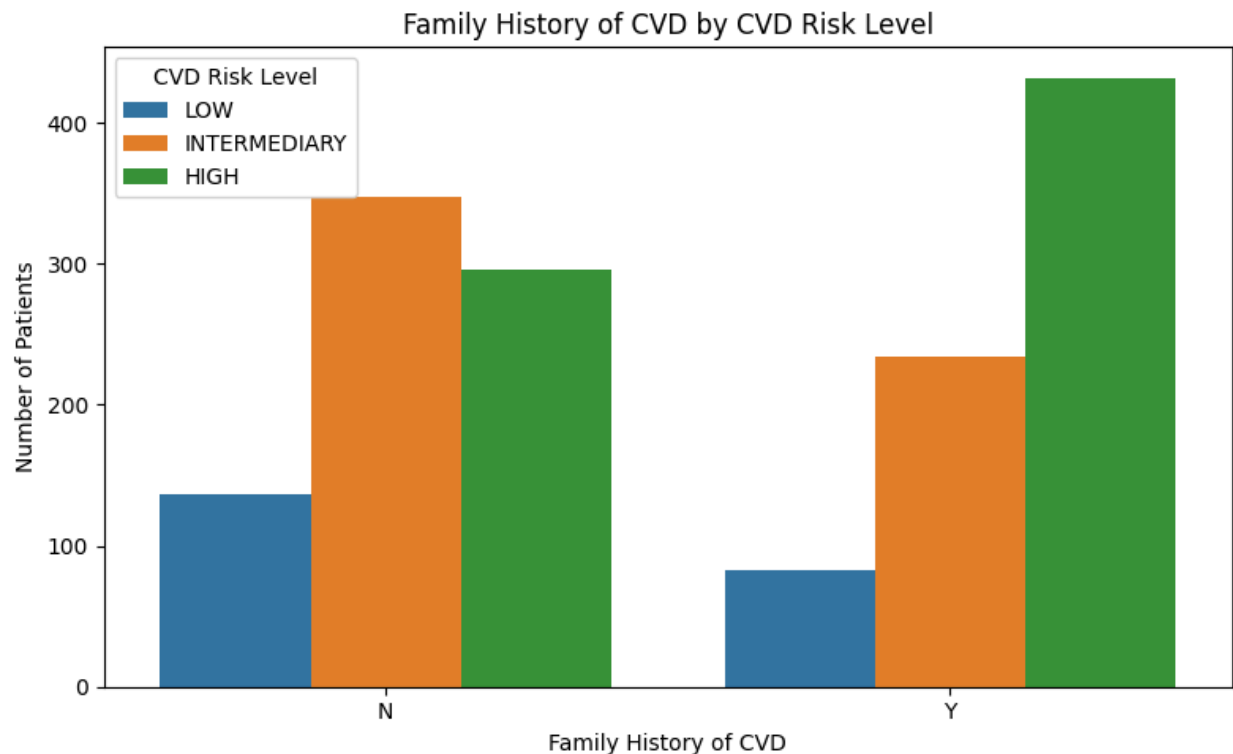


Figure 10: Family History of CVD by Risk Level Countplot

Physical Activity Level by CVD

This countplot compares people who report doing different amounts of physical activity ranging from low to high. The countplot shows that those that have performed low amounts of physical activity have a significantly higher risk of CVD compared to those who performed moderate or high activity. However, there are still patients with moderate and high risk levels despite having higher physical activity indicating that physical activity itself will not suffice as a predictor but it is a strong variable to use in the model when put with other variables.

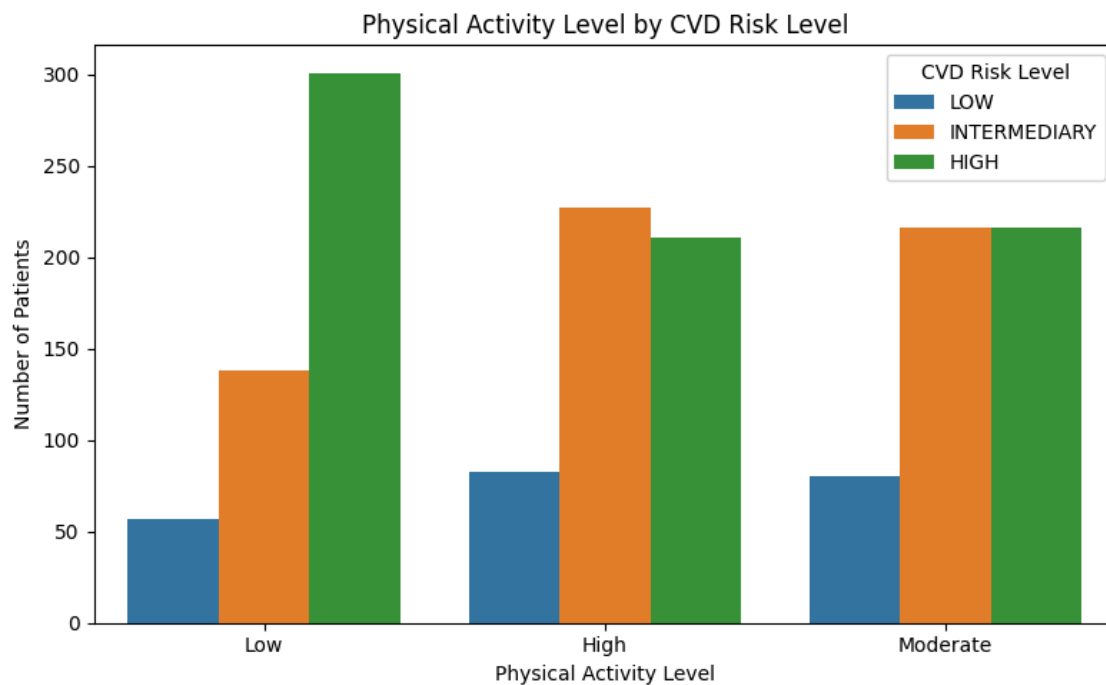


Figure 11: Physical Activity Level by CVD Risk Level Countplot

Blood Pressure Category by CVD Risk Level

This countplot compares all the different blood pressure categories with the risk of getting CVD. They all have the highest values in the “HIGH” category. However, there are also more people with elevated blood pressure which also indicates that CVD is exasperated when blood pressure is higher. However, this alone won’t be a good predictor for the model and simply having the category won’t give us the full scope of everything which is why this needs to be broken down into the blood pressure numbers in order to get an understanding.

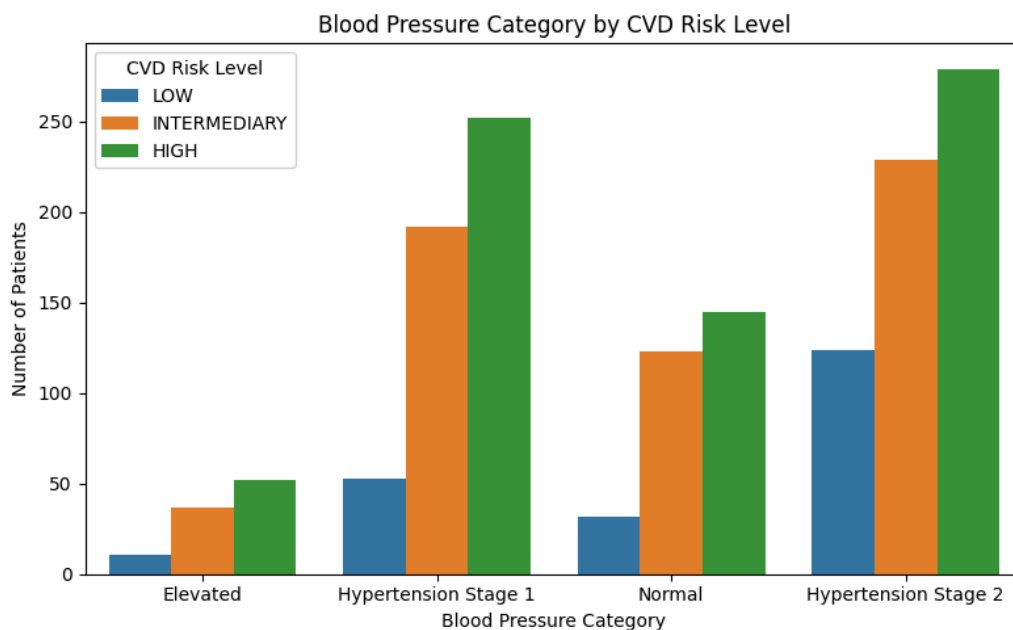


Figure 12: Blood Pressure Category by CVD Risk Level Countplot

Correlation Matrix

To further understand what variables have the strongest correlation and which ones would be best for the model, a correlation matrix needs to be made. This shows what variables have the strongest correlation and if it is suitable for the model. In the correlation matrix, values with a perfect 1.00 score indicates that these two variables are the exact same so it does not count. Some of the variables such as “Estimated LDL (mmHg)” and “Total Cholesterol (mmHg)” & “Abdominal Circumference (cm)” and “Waist-to-Height Ratio” have a correlation of 0.90+ and after further research, is determined that these would be redundant since total cholesterol is made up of estimated LDL and HDL scores whereas waist-to-height ratios are made from abdominal circumferences. BMI is made up of weight and height so BMI would also be redundant however, it is still important as that ratio can give more insight on CVD risks. Finally, height in both centimeters and meters are redundant since they’re the same values so one of them needs to be removed for accurate analysis. Looking at some of the scores, especially the ones being compared to CVD Risk Score have about a 0.50 correlation such as Total Cholesterol, HDL, Estimated LDL, and BMI but as mentioned, some of these values are redundant. Correlation however does not equal causation.

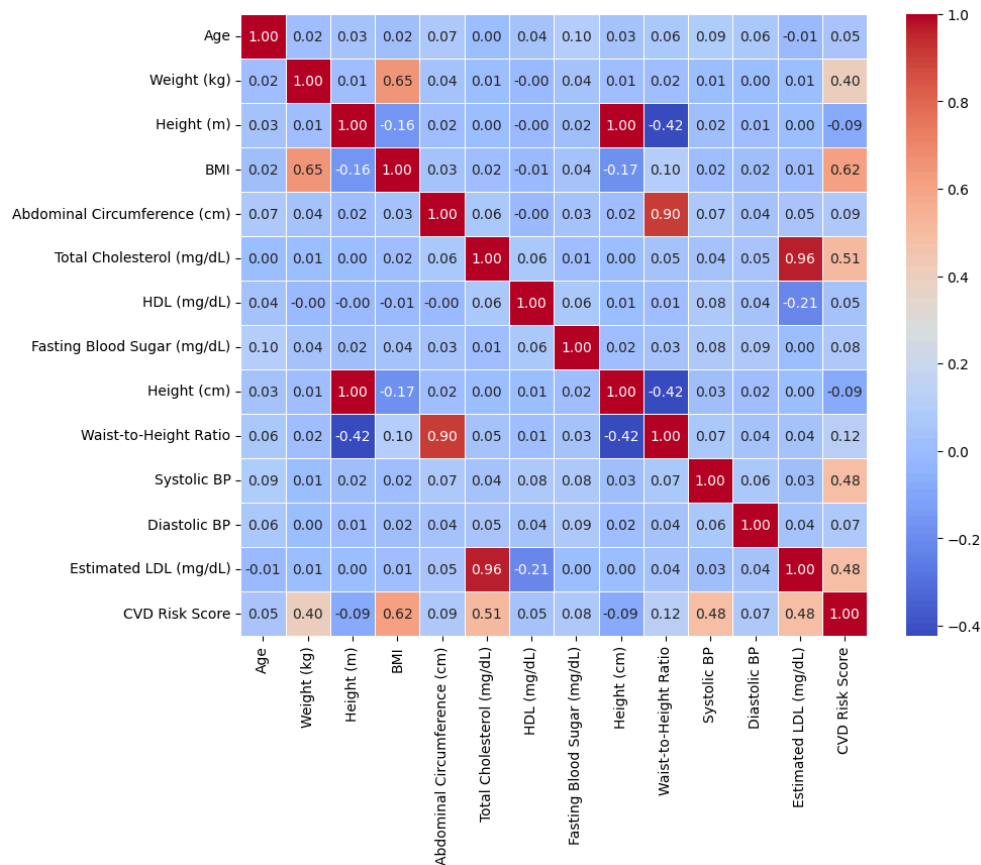


Figure 13: Correlation Matrix

Feature Selection

Feature Selection Criteria

The features were selected based on the correlation matrix, redundancy, statistical analysis, potential data leakage, and model-based importance. Some of the methods that were used for this analysis were the Kruskal-Wallis Analysis and finding strong correlations and determining which ones would be redundant. For the correlations, CVD Risk Level has the highest correlations with estimated LDL, systolic BP, total cholesterol, BMI, and weight (kg). These features will most be kept since they aren't necessarily redundant or repetitive. The only exception is BMI since that is made up of height and weight, however, it has been kept since that ratio tends to work best with tree-based models. To be more deliberate with what features can be used, a Kruskal-Wallis test was performed. Also, the target variable (CVD Risk Level) was dropped from the predictor set so it can be used for model training. The CVD Risk Score, Height (cm), and Blood Pressure (mmHg) were removed from the predictor set since they are redundant variables for the target variable, Height (m), and the systolic & diastolic BP values.

Kruskal-Wallis

The Kruskal-Wallis test (also known as the H-test) is a non-parametric statistical test used to compare three or more independent variables no matter if they are evenly distributed or not [\[10\]](#). This test works better than the ANOVA test because it doesn't assume the dataset is evenly distributed. It can also be used for multiple variables. The dataset used in this project fulfills all of those requirements as it has multiple variables and it is unevenly distributed. Performing the Kruskal-Wallis test led to the results shown below:

	Feature	LOW n	INTERMEDIARY n	HIGH n	H Statistic	p-value	Significant (p < 0.05)
7	Estimated LDL (mg/dL)	728	581	220	67.765648	1.926977e-15	True
3	BMI	726	578	220	57.132564	3.924834e-13	True
6	HDL (mg/dL)	728	581	220	46.045069	1.003322e-10	True
5	Total Cholesterol (mg/dL)	728	581	220	43.626813	3.361692e-10	True
1	Weight (kg)	728	581	220	26.814063	1.504528e-06	True
9	Systolic BP	728	581	220	15.684665	3.927518e-04	True
0	Age	728	581	220	15.228698	4.933216e-04	True
10	Diastolic BP	728	581	220	10.664283	4.833708e-03	True
8	Fasting Blood Sugar (mg/dL)	728	581	220	6.703432	3.502420e-02	True
2	Height (m)	727	577	219	4.905529	8.605537e-02	False
4	Abdominal Circumference (cm)	728	581	220	3.121868	2.099399e-01	False
11	Waist-to-Height Ratio	728	581	220	1.990971	3.695440e-01	False

Table 1: Kruskal-Wallis Test Results with Statistical Significance

This table shows the results of the Kruskal-Wallis test. Each group has approximately 728 LOW values, 581 INTERMEDIATE values, and 220 HIGH values. These groups are not evenly balanced as mentioned before. The H-Statistic shows how different the three groups appear to be for that variable. Larger H-Statistic values generally means there is a stronger evidence for differences. The main thing to look at is the p-value which will determine if the variables are varied across each category or not. If $p < 0.05$, there is a statistically significant difference between at least one pair of groups whereas if $p \geq 0.05$, there is not enough evidence that the groups differ for that specific variable. Almost all of these variables are statistically significant besides height, abdominal circumference, and waist-to-height ratio. This doesn't mean they are important or that they are bad, they just do not have enough evidence to determine the variability across each category. All of this is saying that there are a lot of variables that may vary across categories and can provide useful information for distinguishing between the risk groups.

Post-Hoc Analysis

For significant variables, an FDR correction was performed in order to control the percentage of fake discoveries among all significant results when running multiple multiple tests at once.

	Feature	LOW n	INTERMEDIARY n	HIGH n	H Statistic	p-value	Significant (p < 0.05)	Adjusted p-value	Significant after FDR
7	Estimated LDL (mg/dL)	728	581	220	67.765648	1.926977e-15	True	2.312373e-14	True
3	BMI	726	578	220	57.132564	3.924834e-13	True	2.354900e-12	True
6	HDL (mg/dL)	728	581	220	46.045069	1.003322e-10	True	4.013287e-10	True
5	Total Cholesterol (mg/dL)	728	581	220	43.626813	3.361692e-10	True	1.008508e-09	True
1	Weight (kg)	728	581	220	26.814063	1.504528e-06	True	3.610866e-06	True
9	Systolic BP	728	581	220	15.684665	3.927518e-04	True	7.855036e-04	True
0	Age	728	581	220	15.228698	4.933216e-04	True	8.456943e-04	True
10	Diastolic BP	728	581	220	10.664283	4.833708e-03	True	7.250562e-03	True
8	Fasting Blood Sugar (mg/dL)	728	581	220	6.703432	3.502420e-02	True	4.669894e-02	True
2	Height (m)	727	577	219	4.905529	8.605537e-02	False	1.032664e-01	False
4	Abdominal Circumference (cm)	728	581	220	3.121868	2.099399e-01	False	2.290253e-01	False
11	Waist-to-Height Ratio	728	581	220	1.990971	3.695440e-01	False	3.695440e-01	False

Table 1a: FDR-Corrected Kruskal-Wallis Table

Chi-Squared Analysis

Kruskal-Wallis is great for numerical values, however the categorical values must also be studied to determine which ones are statistically significant and if they can provide useful information for distinguishing between the risk group. In order to do that, the chi-square test must be performed. The chi-square test checks if the categorical values are actually independent or if they depend on each other. This way, data with the weakest link can be removed to drop redundant data and determine which features have the strongest link to the target variable [\[5\]](#).

	Feature	Chi2 Statistic	p-value	Adjusted p-value	Significant after FDR
1	Smoking Status	69.265232	9.104287e-16	5.462572e-15	True
2	Diabetes Status	62.314052	2.942223e-14	8.826669e-14	True
4	Family History of CVD	60.034926	9.195627e-14	1.839125e-13	True
3	Physical Activity Level	50.940082	2.297504e-10	3.446256e-10	True
5	Blood Pressure Category	24.838255	3.658181e-04	4.389817e-04	True
0	Sex	1.381360	5.012350e-01	5.012350e-01	False

Table 2: Chi2 Table

After implementing FDR correction, it is shown that all the categorical variables except for Sex are statistically significant. As mentioned before, this means that these variables are varied enough and are independent from each other and the target variable. This will provide useful information once the models are completed.

Final Feature Set

After completing all the necessary statistical analyses and redundancy checks, these are the final features that will be used in the model:

Numerical

Age

Weight (kg)

Height (m)

BMI

Abdominal Circumference (cm)

Total Cholesterol (mg/dL)

HDL (mg/dL)

Fasting Blood Sugar (mg/dL)

Waist-to-Height Ratio

Systolic BP

Diastolic BP

Estimated LDL (mg/dL)

Categorical

Sex

Smoking Status

Diabetes Status

Physical Activity Level

Family History of CVD

Table 3: Final Feature Set

Machine Learning Methodology

Introduction

Multiple machine learning models were used in order to determine which model would be the most accurate, precise, and effective when it comes to predicting CVD risk. Five models were built and trained: Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, & XGBoost. These models were chosen because they provide variety when it comes to model complexity, variability, & interpretability. Logistic regression was used as a baseline model while the other models were used to evaluate nonlinear and more complex approaches. All of these models were trained using the same training dataset and evaluated using the same testing dataset to ensure consistent, accurate results.

Logistic Regression

Logistic Regression is actually a classification algorithm and is used when there are multi-class variables involved [3]; in our case, our target classifications are “HIGH”, “INTERMEDIARY”, and “LOW”. It is a good baseline model because it is relatively easy to work with and interpret. Establishing a baseline is important because it provides a reference point for determining if more complex models are worth using for predictive performance.

Decision Tree

Decision trees are used for classification and regression tasks. It works like a flowchart and uses step-by-step decisions [2]. It also can potentially identify nonlinear relationships between the predictor variables and CVD risk level. The algorithm recursively divides the observations into smaller groups based on thresholds in the predictor variables. At each split, the algorithm selects a variable and threshold that best separates the target classes. This will continue until a specific stopping condition is reached. Despite being able to capture nonlinear relationships and have easy visualization methods, it can become overly complex and can overfit the training data, so ensemble-based tree models like random forest become useful.

Random Forest

Random Forest is an ensemble learning method that consists of multiple decision trees. Instead of relying on a single decision tree, random forest constructs numerous trees that are made using different subsets of variables. Each tree independently generates a prediction and those are combined to determine the final classification. Random forest is useful because it reduces the complexity of individual decision trees while reducing overfitting of training data. It also provides feature importance measures that can be used to determine which ones contributed the most.

Gradient Boosting

Gradient boosting is another ensemble learning method in which multiple trees are constructed sequentially. Gradient boosting builds each subsequent tree to improve upon the errors made by the existing ensemble. This model progressively reduces prediction errors by giving the most attention to observations that had errors. Gradient boosting is useful since it can capture complex, nonlinear relationships while correcting errors which can achieve stronger predictive performance.

XGBoost

XGBoost is short for Extreme Gradient Boosting which optimizes the error corrections. However, XGBoost incorporates additional optimization techniques which is best for efficiency and reduction of overfitting. XGBoost is useful because it is able to model complex, nonlinear relationships and improve efficiency.

Why These Models?

These models were chosen because they provided a range of modeling techniques that can be compared. Logistic regression is relatively simple and establishes a baseline whereas a decision tree will provide straightforward, nonlinear classification methods. Random forest expands on the decision tree method by combining multiple independently constructed trees. Gradient Boosting and XGBoost combine trees sequentially to progressively improve predictions. Comparing these models allows the study to determine whether increased model complexity and ensemble learning provide improved prediction scores.

Compare the Models

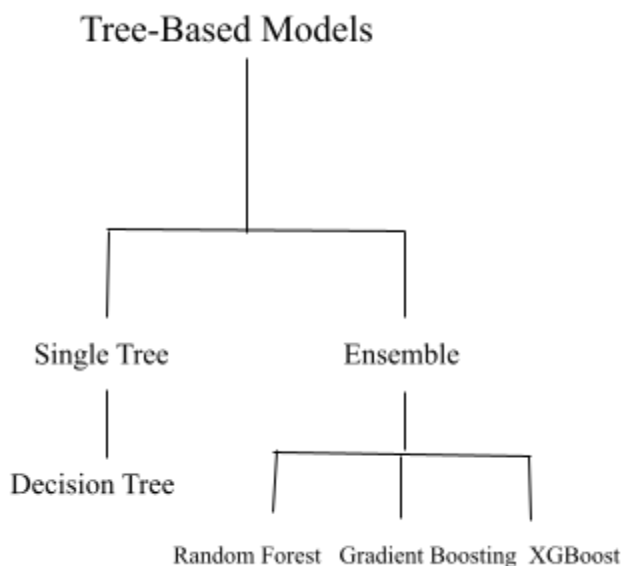


Figure 14: Model Comparison Diagram

Model Training

Training/Testing Split

The dataset was split into two different variables. The X represents the predictor variables while the y represents the target variables. CVD Risk Score and Risk Levels were excluded from the predictor variables to prevent outcome-related information from being used during model training. Then, the data was then split into training and testing sets using an 80:20 ratio. Stratified sampling was applied to maintain a similar distribution of CVD risk categories across both subsets, and a fixed random state of 42 was used to ensure reproducible results.

Feature Scaling

Feature scaling allows the variables to be on a comparable numeric range. First the blood pressure (mmHg) column was dropped since it was redundant to systolic & diastolic BP. Any null values were filled in using the median from the training set. This ensures consistent, accurate results in the end after training.

Model Building (Logistic Regression)

Logistic regression establishes our baseline. Since logistic regression doesn't have a standard method of weighing values, it relies on iterative solvers to gradually reduce errors and find the best solution. Using max_iter defines the upper limit of steps the optimization model can make to find the optimal weights for the model. For this model, max_iter was set to 1000 meaning it would have up to 1000 steps to find the optimal weights for the model.

Decision Tree Training

The decision tree was trained from the pre-processed dataset. It has a maximum depth set to 10 as a base model however, it can be changed in order to determine if more tree depth makes it more precise. It also has min_samples_split which is the minimum samples required to split an internal node. It has the min_samples_leaf which is the minimum samples required to be at a leaf node. From experience, having more tree depth reduces the precision a little bit but slightly increases recall and f1-score. It uses a random state of 42 for reproducibility and uses weighted average for accuracy, precision, recall, F1, and ROC AUC scores.

Random Forest Training

Random forest was trained in a similar way to decision trees. However, the important parameter it uses is the n_estimators parameter which shows how many trees are in the model. It uses a random state of 42 for reproducibility and it is fitted onto a comparable numeric range. Predictions are then made in order to evaluate the model and use weight average for accuracy, precision, recall, F1, and ROC AUC scores.

Gradient Boosting Training

This is also trained using the pre-processed dataset and uses a random state of 42 for reproducibility. It has important parameters such as `learn_rate` which shrinks the step size to prevent overfitting, `max_depth` which is the maximum depth of the individual regression, and then `min_samples_leaf` which is the minimum samples required to be at each leaf node. It is then fitted onto a comparable range and makes predictions on the test set in order to evaluate the model.

XGBoost Training

XGBoost is trained the same way as the other four models. It uses a random state of 42 to ensure reproducible results. It also uses `learning_rate` and `max_depth` just like gradient boosting. Unlike gradient boosting however, it also uses `subsamples` and `colsample_bytree`. `Subsamples` are used for the ratio of the training instance whereas the `colsample_bytree` is used for the ratio of columns when each tree is constructed.

Model Evaluation

Evaluation Metrics

After training the models, the performance of each algorithm was evaluated using different performance metrics. Accuracy, precision, recall, and F1-score were selected since those would provide a detailed assessment of the model performance across the three risk categories. Confusion matrices were also examined to identify patterns in correct and incorrect classifications. Since the distribution of CVD risk categories were not evenly distributed, accuracy alone would not be enough for evaluation. Greater emphasis was placed on precision, recall, and F1-score to determine who effectively each model identified observations within each risk category.

Model Comparison

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	61.76%	60.38%	61.76%	58.80%
Decision Tree	56.21%	55.54%	56.21%	55.82%
Random Forest	61.76%	57.79%	61.76%	57.43%
Gradient	66.01%	61.99%	66.01%	63.29%

Boosting				
XGBoost	66.67%	63.09%	66.67%	64.28%

Table 4: Model Comparison Table

These are all of the weighted averages of the performance metrics. Each result gives very insightful information on what the models are showing us. XGBoost had the highest overall performance with an accuracy of 66.67%, precision of 63.09%, and an F1-score of 64.28%. Gradient Boosting produced the second-highest performance, achieving an accuracy of 66.01% and F1-score of 63.29%, while random forest showed an accuracy of 61.76%. Finally, logistic regression and decision trees are the two models with the lowest performance with accuracy values of 61.76% and 56.21% respectively. These results indicate that ensemble boosting performances, more specifically XGBoost, were more effective at predicting CVD Risk levels within this dataset than the individual decision tree and logistic regression approaches.

Feature Importance

Introduction

Feature importance helps determine if certain features contributed more to the model performance than other features. It also determines if these features are more significant and why they might be more important. Some benefits of feature importance are to help understand why models make certain decisions, to help improve model performance by removing redundant or unimportant features, reduce overfitting by eliminating unnecessary features, and give better data insights by highlighting the most important features.

Random Forest Feature Importance

In the random forest model, it performs the best with a few high performing features. The top five features in this model are Estimated LDL (mg/dL), BMI, Age, HDL (mg/dL), and Total Cholesterol (mg/dL). This indicates that a majority of the model accuracy and precision comes from these features while the others aren't as important and would likely not impact the model much if it was removed.

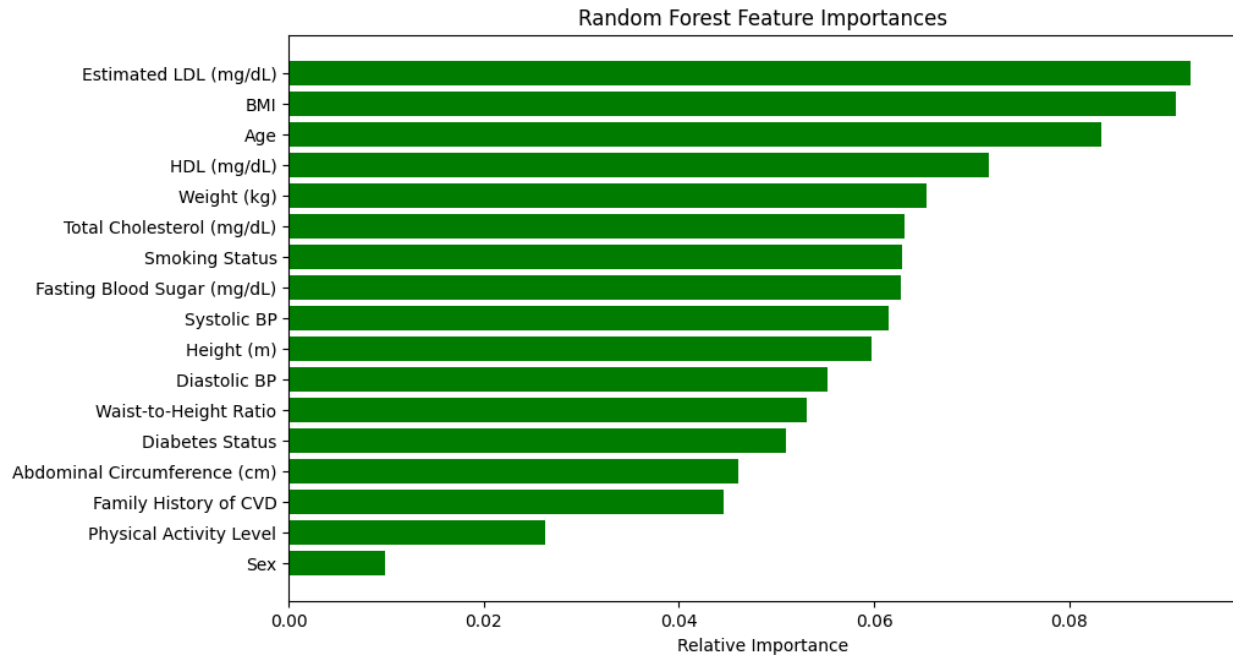


Figure 15: Random Forest Feature Importances Horizontal Barplot

XGBoost Feature Importance

Just like the random forest models, XGBoost only requires a few features to predict risk levels among patients. The top five most important features for XGBoost are Diabetes Status, Family History of CVD, Smoking Status, Estimated LDL (mg/dL), and HDL (mg/dL). This indicates that a majority of the model's precision and accuracy scores come from these few features. The bottom features wouldn't impact the model as much, and in some cases, removing these features may improve the accuracy and precision of the model. For the sake of simplicity and keeping consistent data, these features will remain in the model.

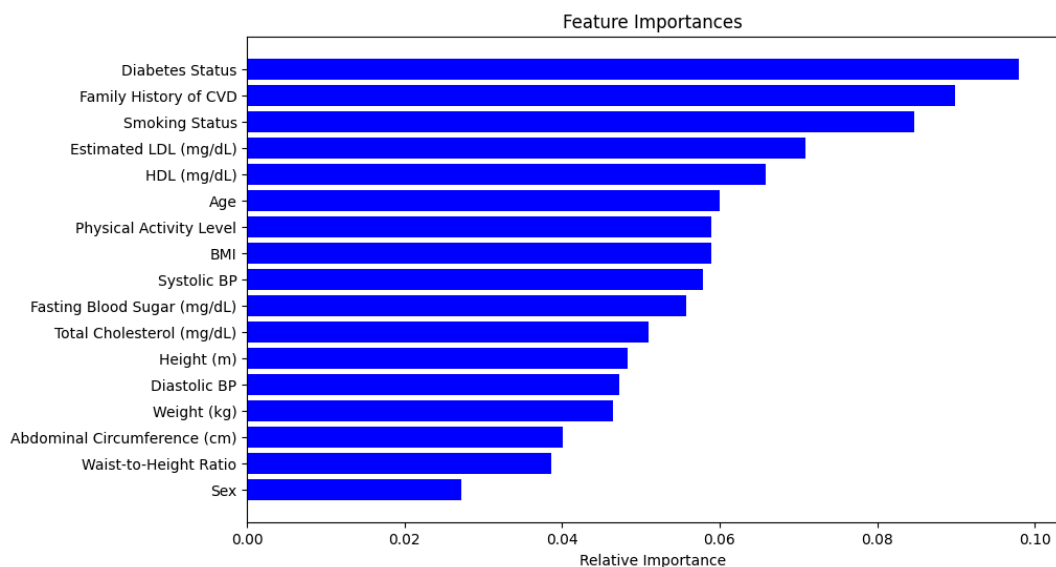


Figure 16: XGBoost Feature Importance Horizontal Barplot

Results

Variables most associated with risk

Looking at both feature importance graphs, the most common variables that were most associated with CVD risk are Estimated LDL, BMI, Age, HDL, & Smoking Status. These features would make sense since South Asians tend to have the highest risk starting from their middle ages alongside being genetically predisposed to having elevated levels of cholesterol levels [\[1\]](#).

Which model performed the best?

The model that performed the best was the XGBoost model. It had an accuracy of 66.67%, a precision score of 63.09%, a recall score of 66.67%, and an F1-score of 64.28%. These numbers indicate that the XGBoost model correctly classified about 67% of all cases. When the model was asked to predict the chances that someone was at risk for cardiovascular disease, it correctly predicted about 63% of those cases. Of all the people who actually had CVD or were at risk for CVD, the model correctly identified about 67% of them. The F1-score combines the precision and recall scores into one measurement. The score 64.28% indicates that there was a moderate balance between the model's ability to correctly identify positive cases and avoid false positives.

Were there any unexpected findings?

Some unexpected findings that were discovered were the fact that a majority of the intermediate & high risk patients started from around the age of 45. Also, groups who had a lower fasting blood sugar still had a significant number of patients that were in the high CVD risk level which further proved that predicting CVD risk would need a combination of factors. One thing that caught significant attention was the fact that the majority of patients fell in the mid-40s category in the age distribution chart which may have some impact on the results.

Differences between the statistical significance and model importance

When comparing the statistical significance with the random forest model importances, the top features of the model importance are pretty similar to the top 5 values of the statistical significant features. In the random forest importances, the top 5 features are estimated LDL, BMI, Age, HDL, and weight whereas the top 5 statistically significant values are estimated LDL, BMI, HDL, Total Cholesterol, & Weight. These values are similar except for the total cholesterol values however total cholesterol is made up of HDL and estimated LDL so it is not completely different.

Discussion

Why were certain features more important?

Certain features were most likely more important due to higher correlation with the target value and hold a stronger, clearer mathematical link to what the model is trying to predict.

Why did certain models perform better?

Certain models performed better most likely due to the distribution of the dataset. As established earlier, the dataset is unequally distributed, meaning that certain models such as random forest or logistic regression will not perform as well as ensemble boosting models. Ensemble boosting models focus on unevenly distributed datasets by automatically fitting the model.

How does this relate to existing research

This project relates to existing research because current research has determined that South Asians may experience CVD risk factors at younger ages and may have a unique cardiovascular risk profile characterized by diabetes, hypertension, and obesity [1]. Machine learning has increasingly been investigated as an alternative or complementary approach to traditional CVD risk assessment due to its ability to identify complex relationships between multiple demographic, clinical, and lifestyle factors. Alongside this, recent research has shown that applying machine learning to CVD risk factors within South Asian settings has become increasingly more popular and common. An example of this comes from Reuter et al. where they applied machine learning to clinical data from individuals in India and Pakistan to predict the likelihood of inadequate CVD risk-factor control, demonstrating the potential for machine learning to support targeted cardiovascular care in South Asian populations [6]. Alongside all of this, this study evaluates using multiple classification algorithms such as logistic regression, decision tree, random forest, gradient boosting, and XGBoost, to predict LOW, INTERMEDIARY, and HIGH CVD risk levels using demographic, clinical, and lifestyle characteristics. Comparing multiple algorithms within a certain population like South Asians in mind allows for a broader investigation of whether machine learning can provide useful information in risk prediction.

Do the results make sense?

In this case, the results do make sense. In current research, the most common causes of CVD among South Asians are hypertension, diabetes, obesity, and sedentary lifestyles [1]. The feature importance graphs have all of these features in the top 5-top 10 categories. Alongside this, the statistically significant variables were estimated LDL, smoking status, BMI, weight, HDL, and total cholesterol.

Potential Implications

The findings of this study suggest that machine learning has the potential to be a powerful supplementary tool for CVD risk prediction among South Asian populations. The ability to classify individuals demonstrates the potential for machine learning to identify patterns across demographic, clinical, and lifestyle characteristics. If it is further developed and trained using larger and more diverse datasets, machine learning-based risk classification can assist healthcare professionals in identifying individuals who may benefit from additional cardiovascular risk assessment or preventive interventions. It can also support earlier identification of elevated risk and encourage greater attention to factors such as diabetes, smoking, physical activity, blood pressure, and lipid levels.

However, the results of this study should not directly diagnose the disease or make decisions. This is because the best performing model achieved an accuracy of 66.67% with an F1-score of 64.28%. This indicates that the model still produces a substantial number of incorrect classifications. In addition, the dataset contains missing values and unequal representation among CVD risk categories, which may impact model performance and generalizability. Additional validation using larger, independent, diverse datasets would be necessary before such models can be considered for real-world healthcare applications.

Limitations

Dataset Limitations

Some limitations in the dataset include a small sample size. This dataset only had 1,529 patients and had a lot of missing/incomplete data. The problem with this is that it is a small section of a larger population. This dataset had lots of missing data which contributed to lower model performance scores than originally anticipated. The data quality itself was pretty decent however this dataset focuses on only one group of South Asians, Bangladeshis. Future datasets must include patients from different countries across South Asia in order to generalize and make accurate predictions.

Model Limitations

The model itself has a lot of limitations. The main one is the class imbalance. As observed earlier, there was a significant imbalance between the different classes with the amount of HIGH risk patients significantly higher than the other two classes. This would impact the model generalizability since this dataset is a small fraction of the population that it is trying to represent. Feature selection is also limited since the only features that were selected were the ones that were recorded for each patient in the original dataset.

Ethical Considerations

There are some pretty significant ethical considerations in this project that must be carefully reviewed. Patient privacy is extremely important as this information is sensitive. This dataset did a good job with keeping patient privacy intact and any dataset that is being used for statistical analysis and machine learning must keep patients' information private to prevent discrimination, bias, and social shame. Alongside this, the dataset, methodologies, and the entire experiment must be unbiased and fair so that no one is discriminated against and the results do not target a single group of people. The population must also be equally represented for accurate results, prevention of discrimination, unbiased discussions, and for proper generalization. Also, predictive models are only used as a complementary tool and are not meant to be used as a method for diagnosis. These models can vary in accuracy and patients may show different symptoms than others. The models also can not be used to generalize an entire population unless the model itself uses population from the entire population or set of populations.

Conclusion

To sum up everything, the problem that was being investigated was the fact that cardiovascular disease is a major health burden, and South Asian populations experience disproportionately higher cardiovascular risks compared to other populations. In order to predict CVD risks and outcomes, a machine learning model was developed and had multiple algorithms created in order to determine if machine learning was viable as a complementary tool for prediction. The dataset was cleaned by finding duplicates, null values, and impossible values. After that, an exploratory data analysis was performed where different variables such as age and cholesterol were compared with the target variable to determine the correlation. After that, a correlation matrix was formed. A set of statistical tests such as Kruskal-Wallis were used to determine the variables that were the most statistically significant against the target variable. After a final feature set was selected, the model was trained with that dataset which was then used to train five algorithms which were logistic regression, decision tree, random forest, gradient boosting, and XGBoost. A model comparison table was created and the best performing model was XGBoost with an accuracy of 66.67% and an F1-score of 64.28%. Finally two feature importance graphs were made and the most important features from those two were estimated LDL, HDL, BMI, Age, Smoking status, Diabetes status, and physical activity level.

Future Work

While this project demonstrates the potential of machine learning for predicting cardiovascular disease (CVD) risk among South Asian populations, several improvements could be explored in future research. A larger and more diverse clinical dataset could improve representation across different South Asian countries, ethnicities, age groups, and socioeconomic backgrounds, allowing the models to better capture differences in CVD risk and improve their generalizability. The models could also be validated using an external dataset to determine whether their performance remains consistent when applied to individuals outside the original dataset. Future studies could compare additional machine learning algorithms and further optimize the current models through hyperparameter tuning, feature engineering, and feature selection. Addressing the class imbalance through techniques such as oversampling, undersampling, SMOTE, or class weighting may also improve the models' ability to identify underrepresented cases. Finally, future research could examine South Asians as more detailed subpopulations rather than as a single group, while incorporating additional clinical, lifestyle, genetic, and socioeconomic factors. These improvements could strengthen the accuracy, reliability, and generalizability of future models and contribute to a more comprehensive understanding of cardiovascular disease risk among diverse South Asian populations.

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Appendix

Appendix A

Variable	Type	Description
Age	Numerical	Patient age
Sex	Categorical	Patient sex
BMI	Numerical	Body mass index
Smoking Status	Categorical	Smoking status
Diabetes Status	Categorical	Diabetes status
Systolic BP	Numerical	Systolic blood pressure
Diastolic BP	Numerical	Diastolic blood pressure
HDL	Numerical	HDL cholesterol
LDL	Numerical	Estimated LDL
CVD Risk Level	Target	Predicted risk category

Appendix B

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1529 entries, 0 to 1528
Data columns (total 22 columns):
#   Column                                     Non-Null Count  Dtype
---  -
0   Sex                                         1529 non-null   object
1   Age                                         1451 non-null   float64
2   Weight (kg)                               1448 non-null   float64
3   Height (m)                                1462 non-null   float64
4   BMI                                         1465 non-null   float64
5   Abdominal Circumference (cm)              1462 non-null   float64
6   Blood Pressure (mmHg)                     1529 non-null   object
7   Total Cholesterol (mg/dL)                 1456 non-null   float64
8   HDL (mg/dL)                               1449 non-null   float64
9   Fasting Blood Sugar (mg/dL)               1462 non-null   float64
10  Smoking Status                             1529 non-null   object
11  Diabetes Status                           1529 non-null   object
12  Physical Activity Level                    1529 non-null   object
13  Family History of CVD                     1529 non-null   object
14  Height (cm)                                1455 non-null   float64
15  Waist-to-Height Ratio                     1450 non-null   float64
16  Systolic BP                               1458 non-null   float64
17  Diastolic BP                              1447 non-null   float64
18  Blood Pressure Category                    1529 non-null   object
19  Estimated LDL (mg/dL)                     1460 non-null   float64
20  CVD Risk Score                            1459 non-null   float64
21  CVD Risk Level                            1529 non-null   object
dtypes: float64(14), object(8)
memory usage: 262.9+ KB
(1529, 22)

```

Appendix C

```
x = df_clean.drop([
    "CVD Risk Level",
    "CVD Risk Score",
    "Height (cm)",
    "Blood Pressure (mmHg)",
    "Blood Pressure Category"
], axis=1)

y = df_clean["CVD Risk Level"]

x_train, x_test, y_train, y_test = train_test_split(
    x,
    y,
    test_size=0.2,
    random_state=42,
    stratify=y
)
```